ScamGuard: Supervised Local Screening of SMS and URLs with Label-Provenance Controls

Implementation-grounded methods draft

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Abstract

We describe supervised local SMS and URL screening with separate classifiers and explicit annotation provenance. Word and character TF-IDF with Logistic Regression classifies SMS; 13 lexical/structural features and character TF-IDF with histogram gradient boosting classify URLs. Fitting uses 3,857 SMS and 9,123 URLs. Validation selects the SMS variant and branch-specific thresholds. Messages with links use the SMS decision and report URL scores separately; no paired fusion or cloud reasoning is deployed. A 39-message English holdout yields TP=24, FN=1, FP=1, TN=13, but all 25 positives belong to one similarity group. The URL source-label holdout yields precision 0.1965 and recall 0.4928; a separate 81-case author-published human audit yields precision 0.6957 and recall 0.7273. Reused Vietnamese diagnostics retain false alarms and missed positives. The tests describe distinct labels and sampling conditions, not a pooled real-world accuracy estimate.

1 Scope and supervised targets

This revision describes the default web model `processed-20261025`. The research question is whether lightweight supervised classifiers can support reproducible local screening under heterogeneous label provenance. Established methods are used; no algorithmic novelty is asserted.

The SMS target follows the project's annotated smishing/scam scope. Non-smishing spam maps to the negative class under the retained annotation policy; negative does not necessarily mean harmless. The URL target is source-reported abuse/phishing versus benign. A message label is not a URL label. The separate human URL audit addresses evidence-supported phishing versus legitimate. The application uses a local Python process. Mobile edge/browser execution, image understanding, dialogue assessment and VLM escalation are not established here.

2 Data and annotation provenance

SMS fitting combines eligible Mishra–Soni v1 records, previously reviewed Vietnamese messages and scope-screened SmishX records. URL fitting uses PhishVN v4/release v3.1.0. Original labels, source record identifiers, hashes and merged lineage accompany exported records.

Mishra–Soni provides 5,971 source-labelled ham/spam/smishing SMS. SmishX provides 1,200 multilingual messages; its authors report manual labels by two researchers with disagreement resolution. Our language/scope screening is AI review, not a new expert verification. It retains 1,017 English scope-eligible records (141 positive, 876 negative), excludes 183 records for language/scope uncertainty, and blocks 421 further candidates linked to reserved groups, including one older training row. The eligible source counts precede grouping, merging and partitioning.

Partition	Train	Validation	Test
SMS records	3857	962	39
SMS positives	193	49	25
URL records	9123	3826	2627
URL positives	1284	321	69
Human-audit URL records	0	0	81
Human-audit URL positives	0	0	22

SMS training contains 3,716 English and 141 Vietnamese records. Provenance is 3,008 source ham/non-smishing spam, 497 source-human records, 350 AI semantic reviews and 2 user scope decisions. Source-human records may merge older duplicates; 497 is not necessarily the number of newly added messages. The corpus is not entirely human or expert verified. The new SMS test is English only.

PhishVN contains 53,116 original URLs. Training admits positive gold and benign gold/silver records, excludes bronze and silver-positive records, and reserves all domains in the 200-case author-published audit. Its 81 evidence-supported phishing/legitimate cases form a separate evaluation. Feed/registry labels do not constitute universal expert verification. No scenario keywords create targets.

The older keyword-derived Combined Labeled Smishing Dataset and 549,346-row URL corpus remain historical sources, not inputs to this fit. The malformed legacy `spam.csv` is excluded as a fallback; the checksum-pinned UCI archive is available only for a separately declared spam-proxy experiment. IMC25 and MIMICS-3500 contribute zero fitted records. No synthetic text–URL pairs are used. Original data, frozen tests and quarantine records are preserved.

3 Supervised method

3.1 Preprocessing

SMS inference uses the frozen `round2_experiment.text_input`: trim trailing dataset URL wrappers/punctuation, apply Unicode NFC and case folding, mask digit runs of at least four digits and collapse whitespace. URL strings remain in SMS input. The older URL/phone/money/OTP masking pipeline is not substituted. NFC does not establish complete homoglyph or obfuscation handling.

URL input uses `external_training.url_input`: supported formatting repairs, host normalization, preserved path/query case, removed fragment and a forced `http` scheme for model input. Originals remain available. Scheme collapsing removes the documented HTTPS collection shortcut; it is not a transport-security recommendation. Invalid/ambiguous source URLs are quarantined. No URL is fetched.

3.2 SMS classifier

The selected representation concatenates word TF-IDF (1–2 grams, at most 10,000 features) and `char_wb` TF-IDF (3–5 grams, at most 16,000 features), both with sublinear term frequency and default per-vectorizer L2 normalization. Vocabularies and inverse-document frequencies are training-fitted only. Logistic Regression uses L2 regularization, $C = 1$, balanced class weights, maximum 1,000 iterations and seed 42:

$$p_{\text{sms}}(t) = \sigma(w^\top \phi_{\text{sms}}(t) + b).$$

For $y_i \in \{0, 1\}$ the weighted cross-entropy is

$$\ell_i = -a_i\{y_i \log p_i + (1 - y_i) \log(1 - p_i)\}, \qquad a_i = v_i \frac{\sum_j v_j}{2 \sum_{j:y_j=y_i} v_j}.$$

Here $v_i = 3$ for Vietnamese records, otherwise 2 for source-human records, otherwise 1. Pinned scikit-learn 1.9.1 computes balanced class weights from sample-weighted class counts. The solver adds its L2 penalty under its $C = 1$ convention. Weights do not certify labels. Signed-label logistic loss would instead require labels in $\{-1, +1\}$.

3.3 URL classifier

Thirteen numeric features are concatenated with `char_wb` TF-IDF (3–5 grams, at most 1,000 features). A training-fitted StandardScaler transforms the numeric block. HistGradientBoostingClassifier uses maximum depth 4, maximum 100 iterations, learning rate 0.1 and seed 42. No 0.80 row subsampling is configured.

Indices	Implemented features
1–3	URL length, hostname length, combined path/query length
4	Count of configured keyword substrings present in the URL
5–6	Configured risk-suffix and trust-suffix flags
7–8	URL digit ratio and ratio of characters in `-_@!~&=?%`
9–10	Hostname hyphens and $\max(0, \text{hostname dots} - 1)$
11	Parsed IP-literal flag, not an IPv4-only regex
12	Brand substring in hostname outside configured official hostname suffixes
13	Shannon entropy of hostname characters

Suffix flags never override learned scores. Approximate subdomain depth is not the registrable-domain grouping rule. Entropy does not prove fraud; no feature ablation establishes its incremental value. Source, tier, status, scenario, split and labels are not model features.

3.4 Branch decision and policy

For substantive text use $p_{\text{sms}}(t)$ with threshold 0.65. Without substantive text, use $\max_{u \in U} p_{\text{url}}(u)$ with threshold 0.22 if URLs exist. Otherwise return unavailable. For the selected branch, $\hat{y} = \mathbf{1}[p \geq \tau]$ uses the unrounded score. Messages containing links report both branches but use the SMS decision. URL characters already in text can contribute to SMS features; an explicitly supplied URL argument is not appended to SMS input.

The displayed rounded 0–100 score uses separate UI bands below 30, 30–79 and at least 80. These are not the binary thresholds or calibrated uncertainty intervals. Scores and midpoint distance are not established real-world fraud probabilities or epistemic uncertainty. Missing, corrupt or incompatible artifacts return unavailable. No paired meta-learner or cloud fallback exists in this fit.

3.5 Historical stacking experiment

The separate 23 October experiment used

$$p_f = \sigma(\beta_0 + \beta_1 p_t + \beta_2 p_u + \beta_3 p_t p_u).$$

Five inner grouped text OOF folds were restricted to training, with fusion evaluated on 2,632 separately held-out linked messages. Interaction F1 was 0.9303 versus 0.9295 for linear stacking; both AUC values were 0.9475 at reported precision. These different data, labels and models do not supply a current-model ablation or comparison to the 39-message test. A positive coefficient is not a significance test. The original draft is retained in `docs/archive/`.

4 Selection and evaluation

SMS grouping connects normalized duplicates, templates, registered domains from a pinned Public Suffix List and character similarity at least 0.80. Prior holdout groups and all published URL-audit domains are blocked from training. Previously fitted groups cannot be reassigned into the new

test. Existing audits find no group or registered-domain overlap across current partitions; complete campaign independence is not established.

The URL branch preserves eligible provider test rows and excludes audit domains and SMS domains known during its original preparation. After SmishX expansion, seven domains occur in both current corpora. Branch-specific partitions remain disjoint and the separately fitted SMS decision does not consume URL scores; however, joint cross-corpus domain independence is not claimed. A future fusion experiment must remove these overlaps before fitting. Since eligible provider validation lacked gold positives, validation includes reserved provider-training groups and eligible provider-validation negatives. No fully temporal SMS evaluation is claimed.

Validation maximizes $F_2 = 5PR/(4P + R)$ subject to false-positive rate at most 0.10, using thresholds 0.05–0.95 in steps of 0.01. SMS candidates are word-only and word+character; URL architecture is fixed and its threshold selected separately. Frozen selections are word+character SMS at 0.65 and URL at 0.22. Saved test results follow frozen selection hashes. The reproduced web fit matches frozen validation scores exactly and fits zero held-out rows. This revision inspects saved predictions; it does not rerun test inference or model selection.

5 Observed results and limits

All evaluations below use their frozen branch-specific binary thresholds.

Evaluation	N	TP	FN	FP	TN	Precision	Recall
New English SMS	39	24	1	1	13	0.9600	0.9600
URL source labels	2627	34	35	139	2419	0.1965	0.4928
Human URL audit	81	16	6	7	52	0.6957	0.7273
Reused SMS diagnostic	1164	29	9	6	1120	0.8286	0.7632
Reused Vietnamese	70	14	3	9	44	0.6087	0.8235

The new English test has F1=0.9600 but just 11 similarity groups; all positives are in one group. It does not establish cross-campaign or Vietnamese robustness. URL source-test F1=0.2810 and human-audit F1=0.7111 concern distinct targets and sampling conditions; neither test is discarded or pooled.

On the same reused 70-message Vietnamese diagnostic, the previous external model had TP=17, FN=0, FP=20, TN=33. The current model reduces false alarms to 9 but misses 3 positives (F1=0.7000). Reused diagnostics are not new independent tests or threshold-tuning data. No significance, campaign-level confidence interval, neural-model superiority or deployment readiness is asserted.

6 Runtime and remaining evidence

The adapter verifies bundle SHA-256, frozen helper hashes, estimator runtime and thresholds. API execution starts Python per request; import and model-loading costs belong in endpoint latency. No end-to-end mobile/edge timing, cloud comparison, measured API saving or calibrated VLM routing is established. The comparison PDF's sub-4ms and 74.2% saving claims are not current findings.

Further work needs independently human-verified Vietnamese labels, broader positive campaign coverage and provenance-specific evaluation. Feature ablations and a new fusion study need newly reserved evaluation data. A future fusion experiment must use actual paired messages, message-level targets and training-only grouped OOF scores. Calibration, VLM escalation and mobile latency remain uncompleted studies.

7 Reproduction and sources

See `ml_model/README.md`, the current-fit report in `processed_agent_fit_2026-10-25/`, frozen SMS/URL experiment reports and `docs/web-model-activation-2026-10-25.md`. The read-only manifest audit and comparison are `docs/paper-alignment-audit-2026-10-25.json` and `docs/paper-framework-comparison-2026-10-25.md`. This methods draft still needs a source-verified Introduction/Related Work, author details and the official conference template before submission.

- https://data.mendeley.com/datasets/f45bkkt8pr/1.
- https://github.com/yizhu-joy/SmishX/tree/116a8c827741e0572563f678d25ed04306b1e3ff.
- https://data.mendeley.com/datasets/b97hxbxtpd/4.
- https://scikit-learn.org/stable/modules/cross_validation.html.
- https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression.
- https://scikit-learn.org/stable/modules/calibration.html.